NM4QM 2026 · HANDS-ON SESSION · 45 MIN

Spin and anomalous Hall effects with WannierBerri and GPAW

Symmetry-adapted Wannier functions, spin–orbit coupling on demand, the spin Hall effect of graphene and the anomalous Hall effect of the altermagnet MnTe



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THE STORY

Two puzzles, two acts

0–8 min	8–19	19–37	37–45
Setup and ideas	Act 1: graphene	Act 2: MnTe	Your material, questions

TWO PUZZLES

- Spin–orbit coupling turns graphene into a quantum spin Hall insulator with a gap of 42 μeV
- MnTe has no net magnetisation, yet shows an anomalous Hall effect

WHAT YOU WILL DO

- Build symmetry-adapted Wannier functions straight from GPAW
- Switch spin–orbit coupling on, off and around, without new DFT
- Compute spin and anomalous Hall conductivities

Kane, Mele, PRL 95, 226801 (2005) · Gonzalez Betancourt et al., PRL 130, 036702 (2023)

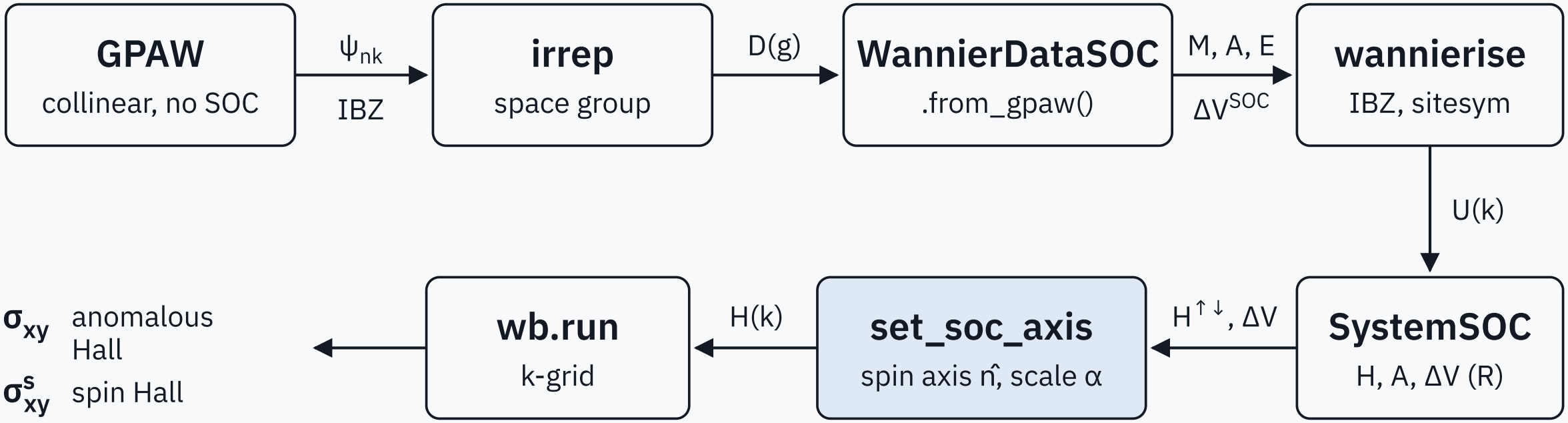
SETUP

Before we start: one check, 20 seconds

```
cd wannier-berri-demo && conda activate wbdemo
python setup/check_setup.py        # should end with "Ready."
cd 01_graphene
python 1_dft.py                    # 10 s
jupyter lab graphene.ipynb
```

- No environment yet: `bash setup/setup.sh` (15 min)
- DFT fails: `tar -xzf dft-files.tar.gz` in `wannier-berri-demo/`
- Fell behind: `bash run_all.sh` (6 min); finished notebooks in `reference/`
- Numbers on the slides are checkpoints: compare them with yours

One Python pipeline, GPAW to conductivities



Each box is one call in the notebooks. The shaded step needs no new DFT and no new Wannierisation.

$\psi_{n\mathbf{k}}$: Bloch states · $D(g)$: symmetry matrices on them · M, A, E: overlaps, projections, energies · ΔV^{SOC}: SOC matrix elements from the PAW spheres · $U(\mathbf{k})$: Wannier gauge · H, A, $\Delta V(\mathbf{R})$: Hamiltonian, Berry connection and SOC in the Wannier basis · $\hat{\mathbf{n}}$, α: spin axis, SOC scale · σ_{xy}: anomalous Hall (charge current), σ^{s}_{xy}: spin Hall (spin current) conductivity

Berry curvature needs millions of k-points; DFT gives tens

Intrinsic anomalous Hall conductivity:

$$\sigma_{xy} = -\frac{e^2}{\hbar} \int_{\mathrm{BZ}} \frac{d^3k}{(2\pi)^3} \sum_n f(\varepsilon_{n\mathbf{k}})\, \Omega_n^z(\mathbf{k})$$

$$\Omega_n^z = -2\,\mathrm{Im} \sum_{m \neq n} \frac{v_{nm}^x\, v_{mn}^y}{(\varepsilon_n - \varepsilon_m)^2}$$

f: occupation, ε_n: band energy, $v_{nm}^a = \langle u_n | \partial_{k_a} H | u_m \rangle$. Ω_n^z is large and narrow wherever two bands come close.

Wannier functions give $H(\mathbf{k})$ at any $\mathbf{k}$:

$$H_{ij}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}}\, \langle w_{i\mathbf{0}} | \hat{H} | w_{j\mathbf{R}} \rangle$$

$w_{j\mathbf{R}}$: Wannier function j in the cell at $\mathbf{R}$; a few hundred $\mathbf{R}$ are enough.

Symmetry-adapted Wannier functions: DFT on irreducible k only

Each Wannier function should transform like its orbital:

$$\hat{g}\, w_{i\mathbf{R}}(\mathbf{r}) = \sum_j D_{ji}(g)\, w_{j,\, g\mathbf{R}}(\mathbf{r})$$

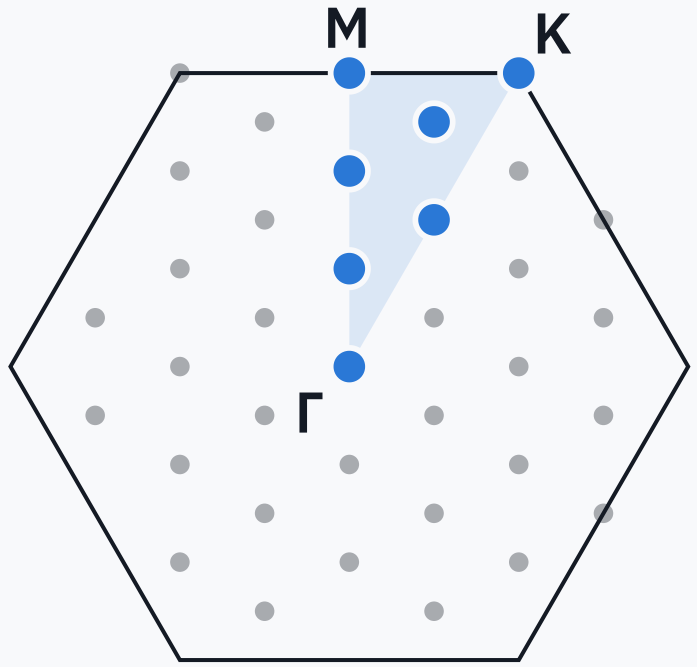
$\hat{g}$: symmetry operation, $D(g)$: its matrix on the orbitals of one site

Then the gauge at g**k** follows from **k** (Sakuma 2013):

$$U(g\mathbf{k}) = \tilde{d}(g,\mathbf{k})\, U(\mathbf{k})\, D(g,\mathbf{k})^{\dagger}$$

U: Bloch → Wannier gauge, $\tilde{d}$: g acting on the Bloch states (from irrep)

- DFT on 7 of 36 k-points for graphene, 24 of 144 for MnTe
- Degeneracies come out exact



6 × 6 grid: 36 points, 7 irreducible (blue)

METHODS

Wannierisation: WannierBerri vs Wannier90

$$\Omega = \sum_n \left[\langle w_n | r^2 | w_n \rangle - \left| \langle w_n | \mathbf{r} | w_n \rangle \right|^2 \right] = \Omega_{\mathrm{I}} + \tilde{\Omega}$$

Ω: spread of the Wannier functions · Ω_{I}: fixed by the chosen subspace (disentanglement) · $\tilde{\Omega}$: depends on the gauge $U(\mathbf{k})$ (localisation)

	Wannier90	**WannierBerri, this demo**
Input	.amn, .mmn, .eig files from a DFT interface	computed in Python from the GPAW wave functions
k-points	full grid; irreducible ones with site_symmetry	irreducible ones
Algorithm	disentangle first, then minimise $\tilde{\Omega}$ by steepest descent or conjugate gradients	one loop: a disentanglement step and a local gauge update per iteration, no gradient descent
Symmetry	site_symmetry with a .dmn file from pw2wannier90, not with a frozen window	from irrep, for any code it reads; frozen window allowed
SOC	spinor DFT run with SOC	collinear run, SOC added afterwards; spin ↓ from symmetry in altermagnets

Marzari, Vanderbilt, PRB 56, 12847 (1997) · Souza, Marzari, Vanderbilt, PRB 65, 035109 (2001) · Sakuma, PRB 87, 235109 (2013) · Pizzi et al., JPCM 32, 165902 (2020)

METHODS

Post-processing: WannierBerri vs postw90

$$H(\mathbf{K}+\boldsymbol{\kappa}) = \sum_{\mathbf{R}} e^{i\boldsymbol{\kappa}\cdot\mathbf{R}} \left[e^{i\mathbf{K}\cdot\mathbf{R}}\, H(\mathbf{R})\right]$$

$\boldsymbol{\kappa}$: points of a coarse FFT grid · $\mathbf{K}$: shifts of that grid; one FFT per shift, the shifts together give the dense grid

	postw90	**WannierBerri**
$H(\mathbf{R}) \to H(\mathbf{k})$	sum over **R** at every k, cost $\propto N_{\mathbf{R}}$	FFT over sub-grids, cost $\propto \log N_{\mathrm{FFT}}$ per k
Symmetry	full grid	irreducible points of the magnetic point group, result symmetrised
Hot spots	finer mesh where |Ω| exceeds a threshold	recursive adaptive refinement
Fermi levels	cost grows with their number	~1000 at almost no extra cost
SOC axis	fixed by the DFT run	`set_soc_axis` : any axis and strength

Up to three orders of magnitude faster on dense grids. In this demo: 1401 Fermi levels in one run; for $\hat{\mathbf{n}} \parallel y$ only the k-points irreducible under m′m′m (8 operations) are computed.

Wang, Yates, Souza, Vanderbilt, PRB 74, 195118 (2006) · Tsirkin, npj Comput. Mater. 7, 33 (2021) · Pizzi et al., JPCM 32, 165902 (2020)

Does graphene have a gap?

Step	Run	Checkpoint
0 · DFT	`python 1_dft.py`	10 s, `gpaw/nscf.gpw`
1 · Symmetry	irrep	48 operations, 7 of 36 k-points
2 · Wannier functions	`wannierise`	two equal p_z-like functions
3 · Bands	interpolation	blue on grey in the window
4 · SOC on demand	`set_soc_axis`	gap ≈ 36 μeV (GPAW ≈ 39 μeV)
5 · Spin Hall	`wb.run`	plateau at e/2π, 30 s

Graphene: DFT in 10 s, two p_z Wannier functions

- PBE, 400 eV, 6×6×1 grid, 15 Å vacuum, no SOC
- NSCF only on the 7 irreducible k-points
- Frozen window E_F − 2.9 … E_F + 2.8 eV, plus the π state at Γ

2
equal, compact p_z functions

48
symmetry operations

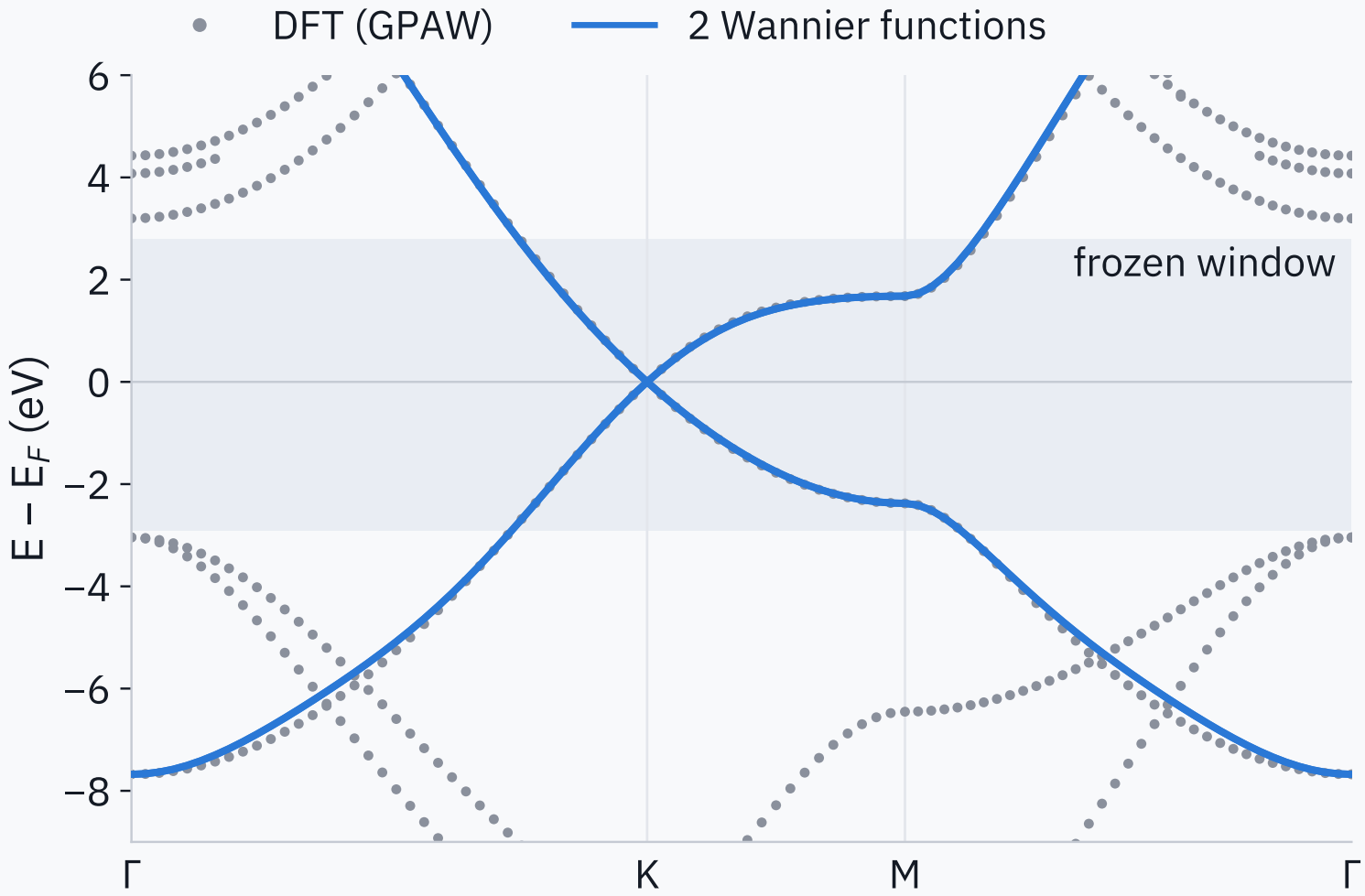
3 s + 2 s
from_gpaw + wannierise

```
calc = GPAW(mode=PW(400), xc="PBE",
            kpts={"size": (6, 6, 1), "gamma": True})
atoms.calc = calc
atoms.get_potential_energy()
sg = SpaceGroup.from_gpaw(calc)                    # irrep
kpts = sg.get_irreducible_kpoints_grid((6, 6, 1))  # 7 of 36
calc.fixed_density(kpts=kpts, nbands=20).write(
    "gpaw/nscf.gpw", mode="all")

# graphene.ipynb
pz = Projection(position_num=[[1/3, 2/3, 0], [2/3, 1/3, 0]],
                orbital="pz", spacegroup=sg)
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=ProjectionsSet([pz]))
wandata.wannierise(froz_min=EF - 2.9, froz_max=EF + 2.8,
                   frozen_states={0: [1]},  # pi at Gamma
                   num_iter=100, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata, berry=True)
```

Two Wannier bands follow DFT through the frozen window

- Grey: GPAW along Γ–K–M–Γ, not used in the fit
- Blue: the two-band Wannier model
- Shaded: frozen window



SOC as a matrix you can rotate and scale

$$H(\mathbf{k}) = H_0(\mathbf{k}) \otimes \sigma_0 + \alpha \sum_c \Delta V^c(\mathbf{k}) \otimes \sigma_c(\hat{\mathbf{n}})$$

$$\Delta V^c_{mn}(\mathbf{k}) = \sum_{a,ij} \langle \tilde{\psi}_{m\mathbf{k}} | \tilde{p}^a_i \rangle \, \langle \phi^a_i | \, \xi_a(r) \, L_c \, | \phi^a_j \rangle \, \langle \tilde{p}^a_j | \tilde{\psi}_{n\mathbf{k}} \rangle$$

$$\xi_a(r) = \frac{1}{4m^2c^2} \frac{1}{r} \frac{dV_a}{dr}$$

H_0: Wannier Hamiltonian without SOC · $\sigma_c(\hat{\mathbf{n}})$: Pauli matrices, z along the spin axis $\hat{\mathbf{n}}$ · α: SOC scale, 1 = real · $\tilde{\psi}, \tilde{p}^a_i, \phi^a_i$: PAW pseudo wave functions, projectors, partial waves of atom a · L_c: orbital angular momentum · V_a: potential in sphere a

- Same PAW matrix elements as GPAW's own SOC
- New $\hat{\mathbf{n}}$ or α: no new DFT

```
K = [1/3, 1/3, 0]
energy = wb.calculators.tabulate.Energy(
    degen_thresh=1e-8)
for alpha in [0, 1]:
    system.set_soc_axis(alpha_soc=alpha)
    E = wb.evaluate_k(system, k=K,
            calculators={"E": energy})
```

```
             Dirac point     gap at K
alpha = 0    +29 ueV          0
alpha = 1    +29 ueV         36 ueV
GPAW SOC     +28 ueV         39 ueV
```

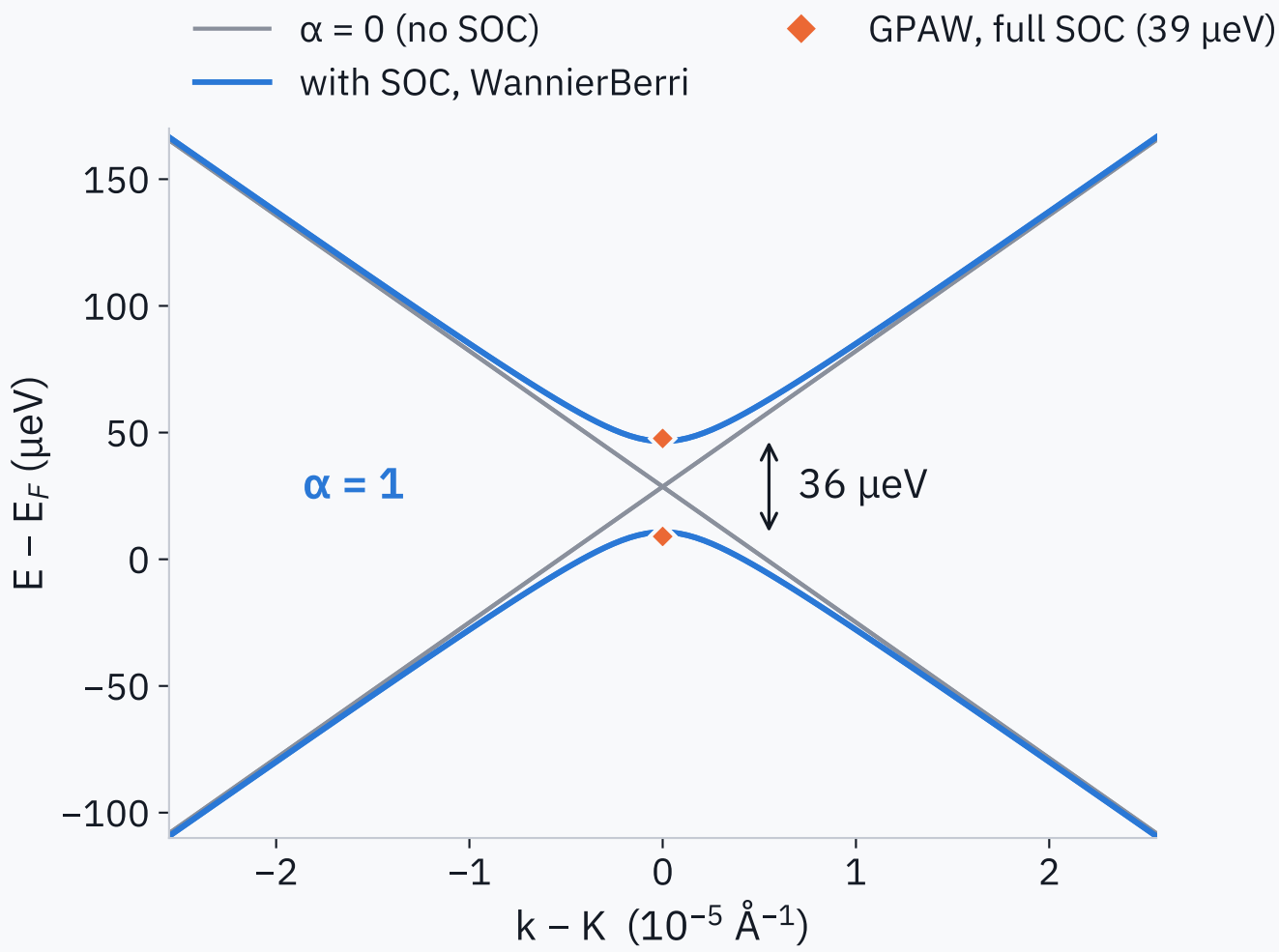
`degen_thresh=1e-8` : the default 10^{-4} eV would merge the split pair

Kane–Mele gap: 36 µeV from two Wannier functions

SOC gap at the Dirac point K

Wannier model, two p_z	**36 µeV**
GPAW, all 20 bands	**39 µeV**
Experiment (Sichau 2019)	**42 µeV**

- p_z has $m = 0$: the gap comes from its small d part in the PAW spheres (Gmitra 2009)
- The model has SOC to first order; the 3 µeV it misses come through the σ bands

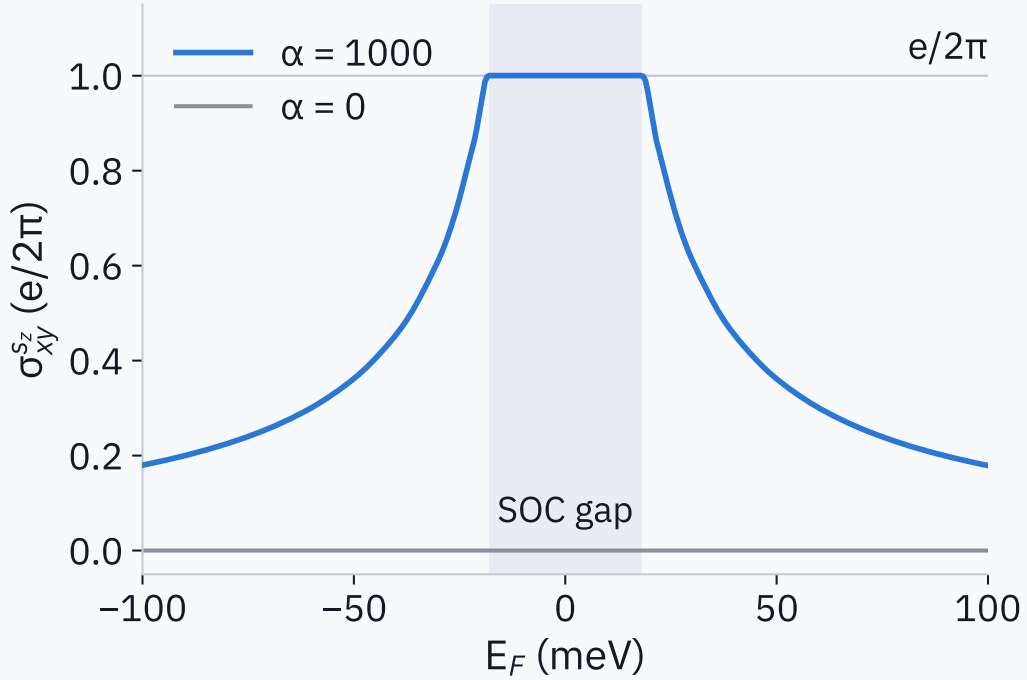


Kane, Mele, PRL 95, 226801 (2005) · Gmitra et al., PRB 80, 235431 (2009) · Sichau et al., PRL 122, 046403 (2019)

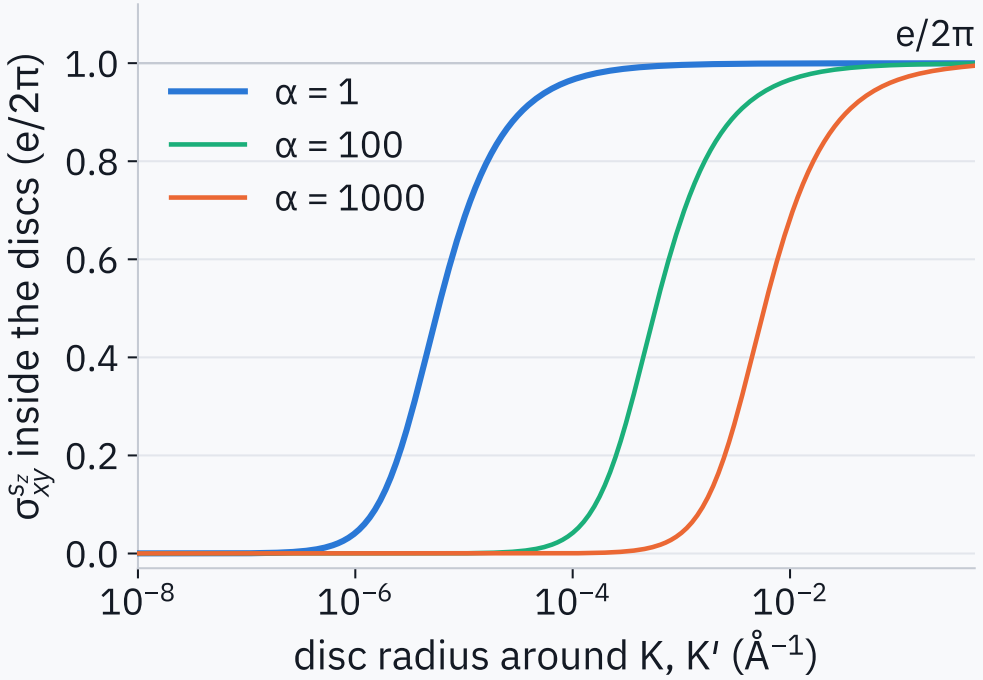
Spin Hall conductivity: e/2π, also at the real gap

$$\sigma_{xy}^{s_z} = \frac{e}{\hbar}\int \frac{d^2k}{(2\pi)^2}\sum_n f_{n\mathbf{k}}\,\Omega_n^{s_z}, \qquad \Omega_n^{s_z} = -2\hbar^2\,\mathrm{Im}\sum_{m\neq n}\frac{(j_x^{s_z})_{nm}(v_y)_{mn}}{(\varepsilon_n-\varepsilon_m)^2}$$

$j_x^{s_z} = \frac{1}{2}\{s_z, v_x\}$: spin current, v: velocity. With s_z conserved, $\sigma_{xy}^{s_z} = (e/2\pi)(C_\uparrow - C_\downarrow)/2 = e/2\pi$



α = 1000 (gap 36 meV): plateau at e/2π



α = 1, the real 36 μeV gap: also e/2π

ACT 2 · MNTE.IPYNB

A Hall effect without magnetisation

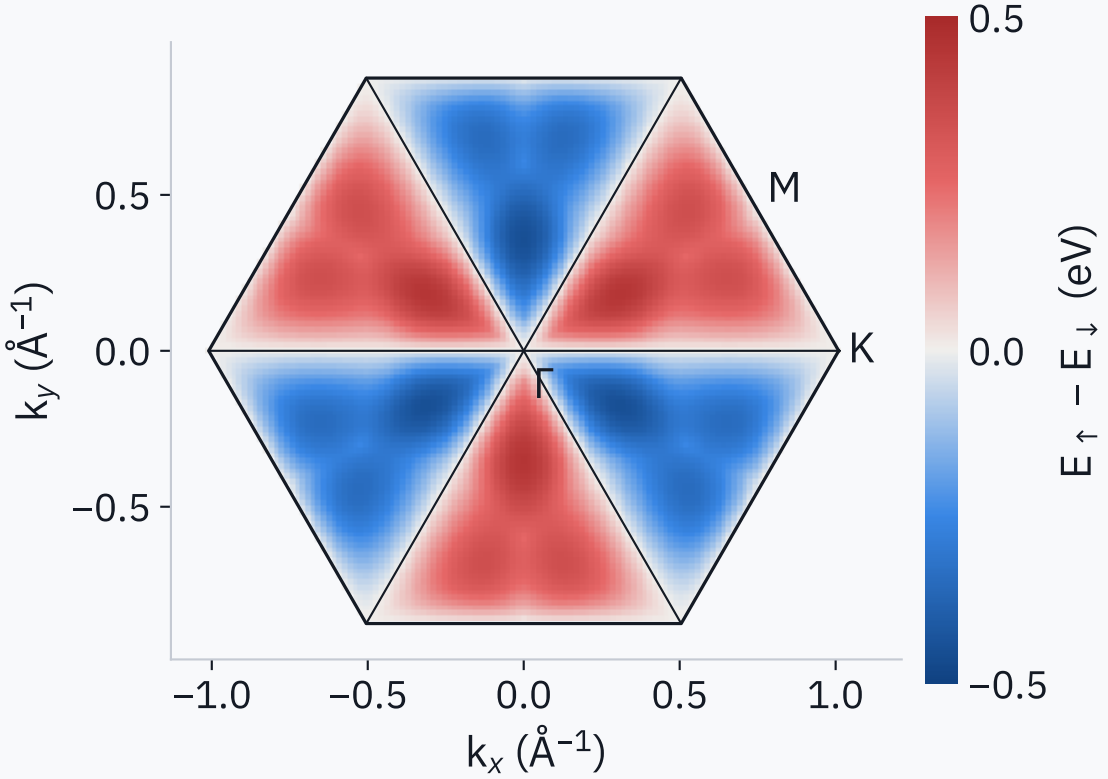
Step	Run	Checkpoint
0 · DFT	`python 1_dft.py`	45 s, moments ≈ ±4.4 μ_B
1–2 · Spin ↑ Wannier	`altermagnetic=True`	24 of 144 k-points, 25 s
3 · Splitting	no SOC	up to ≈ 1 eV at $k_z \neq 0$
4 · Turn $\hat{\mathbf{n}}$	`set_soc_axis`	SOC shift at A 4–5× smaller in plane
5 · Anomalous Hall	three $\hat{\mathbf{n}}$	0 for x and c, up to ≈ 80 S/cm for y
6 · Rotate in plane	0–30°	far from sin 3φ

Altermagnets: zero net moment, spin-split bands without SOC

- Two sublattices related by a rotation or mirror, not by a translation or inversion
- Bands spin split without SOC; the sign alternates in **k** (d-, g-, i-wave)
- No PT or *t*T symmetry, so with SOC an anomalous Hall effect is possible, depending on the Néel vector
- **MnTe**: NiAs type, $T_N \approx 310$ K, gap ≈ 1.3 eV

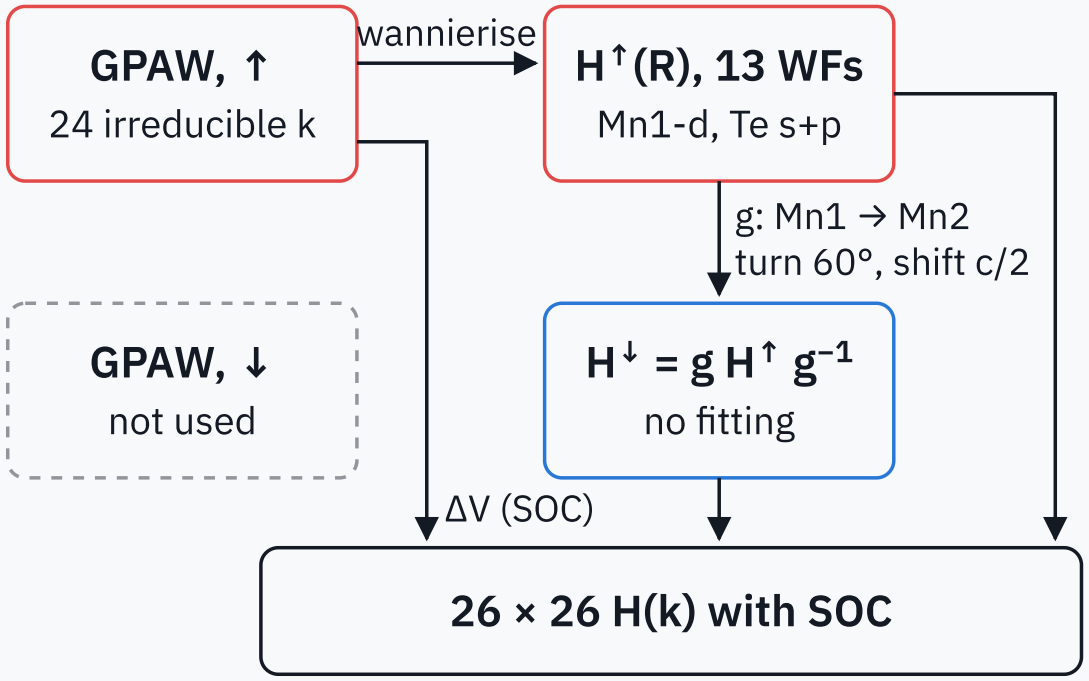
$$[C_2 \parallel g]: \qquad E_\uparrow(\mathbf{k}) = E_\downarrow(R_g\mathbf{k})$$

C_2: flips all spins, g: swaps the Mn sublattices, R_g: its rotation part



MnTe, no SOC: $E_\uparrow - E_\downarrow$ at $k_z = 0.35$ Å^{-1}

Wannierise spin ↑ only, spin ↓ follows by symmetry

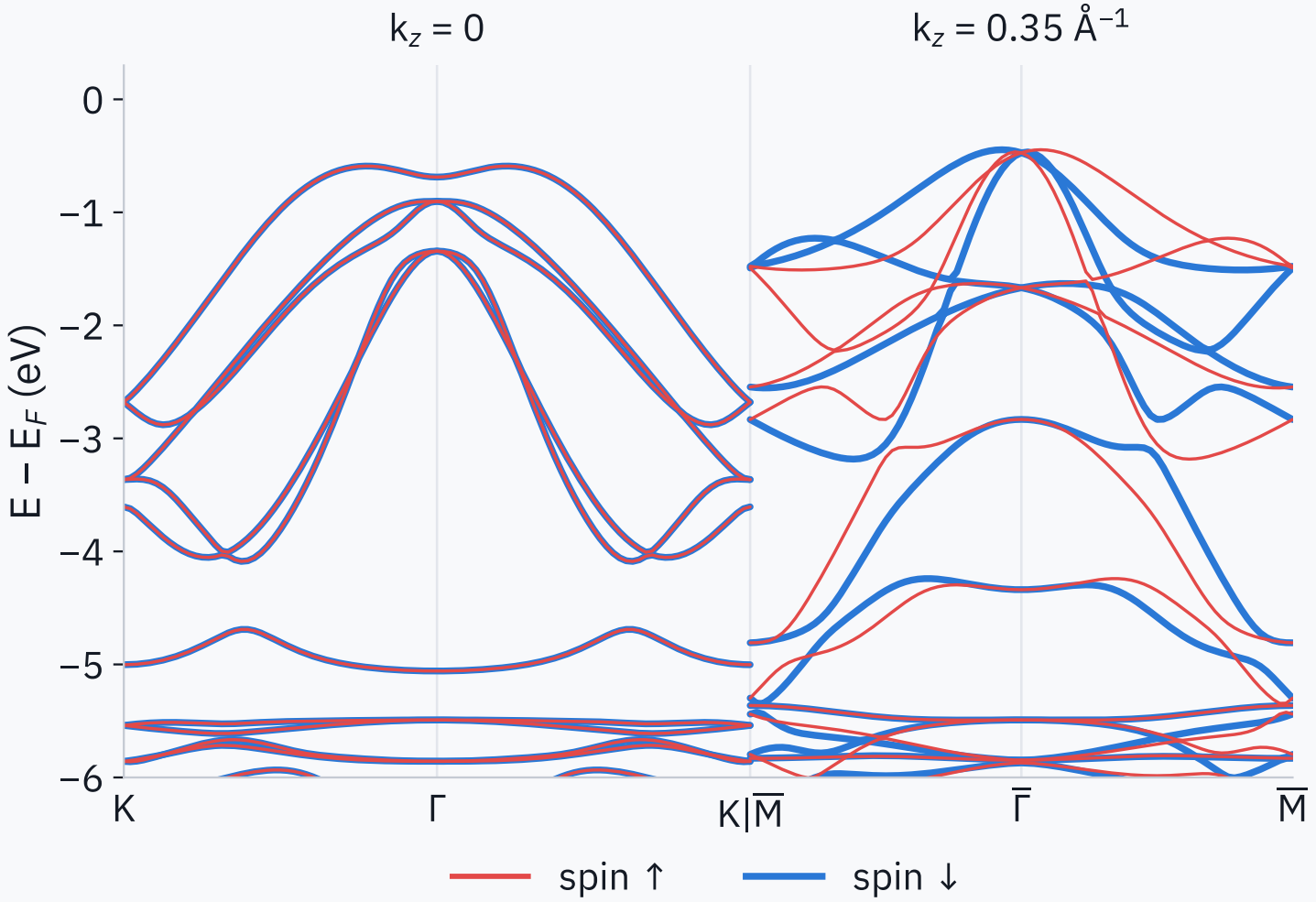


PBE+U (U = 4 eV): moments ≈ ±4.4 μ_B, gap ≈ 0.8 eV (exp. 1.3–1.4 eV; a larger U does not fix it)

```
calc = GPAW("gpaw/nscf.gpw")   # PBE+U, 24 of 144 k
sg = SpaceGroup.from_gpaw(calc)
Te = [[1/3, 2/3, 1/4], [2/3, 1/3, 3/4]]
projections = ProjectionsSet([
    Projection(position_num=[0, 0, 0], orbital="d",
               spacegroup=sg),          # Mn1 only
    Projection(position_num=Te, orbital="sp2",
               spacegroup=sg, xaxis=[0, -1, 0],
               rotate_basis=True),
    Projection(position_num=Te, orbital="pz",
               spacegroup=sg)])
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=projections,
    altermagnetic=True,   # spin down from symmetry
    IBstart=8)            # skip Mn 3s, 3p
wandata.wannierise(froz_min=-np.inf,
                   froz_max=EF + 0.1,
                   num_iter=200, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata,
                                       berry=True)
```

No SOC: degenerate at $k_z = 0$, split by up to 1 eV at $k_z \neq 0$

- Red: spin ↑, Wannierised
- Blue: spin ↓, from symmetry
- At $k_z = 0.35$ Å^{-1} up to ≈ 1 eV apart, with opposite order on the two sides of $\bar{\Gamma}$: g-wave



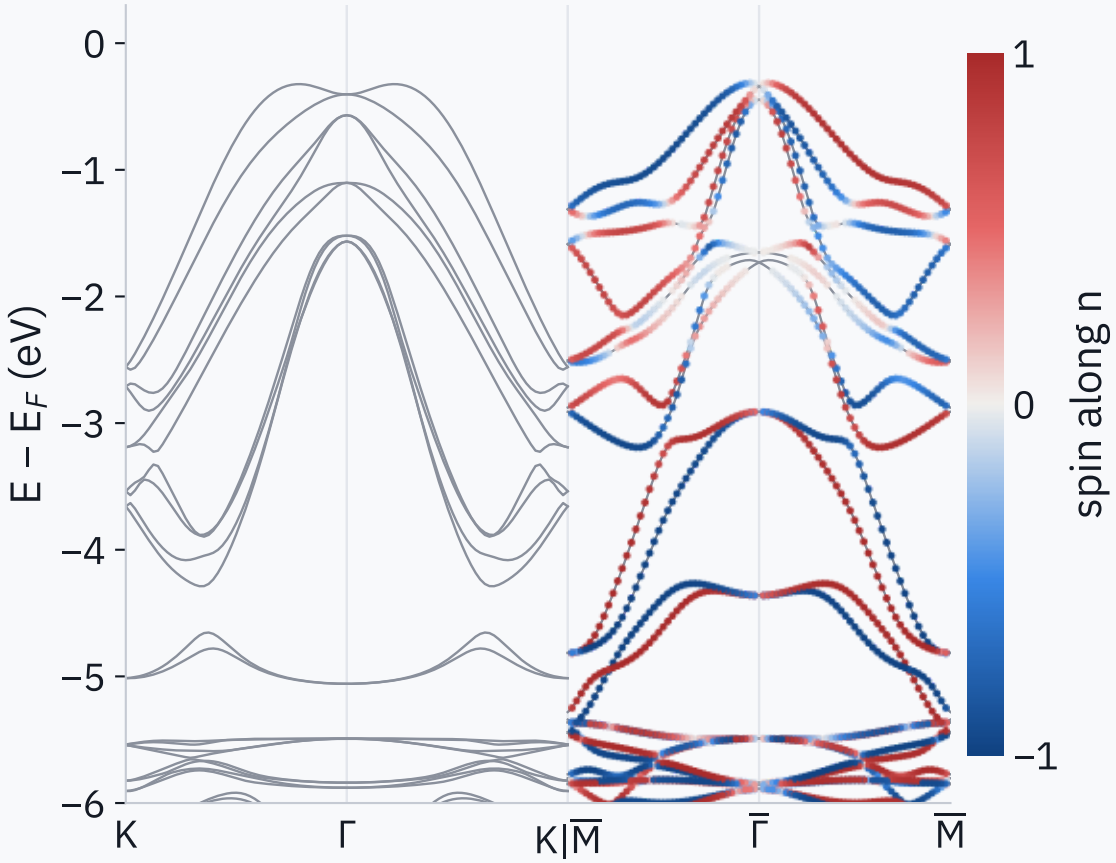
With SOC the Néel vector picks the magnetic point group

Néel vector n̂	Group	Hall vector
∥ c	6′/m′m′m	none
∥ a = x	mmm	none
⊥ a = y	m′m′m	$\sigma_{xy} \neq 0$

$$g: \ \boldsymbol{\sigma}_{\mathrm{H}} \to \det(R_g)\, R_g\, \boldsymbol{\sigma}_{\mathrm{H}}$$
$$\mathcal{T}: \ \boldsymbol{\sigma}_{\mathrm{H}} \to -\boldsymbol{\sigma}_{\mathrm{H}}$$

$\boldsymbol{\sigma}_{\mathrm{H}} = (\sigma_{yz}, \sigma_{zx}, \sigma_{xy})$: Hall vector · R_g: rotation part of g · $\mathcal{T}$: time reversal · primed: combined with $\mathcal{T}$

SOC shift of the top level at A: ≈ 0.3 eV for **n̂** ∥ c, 4–5 times smaller for **n̂** in plane; the model agrees with GPAW's full SOC

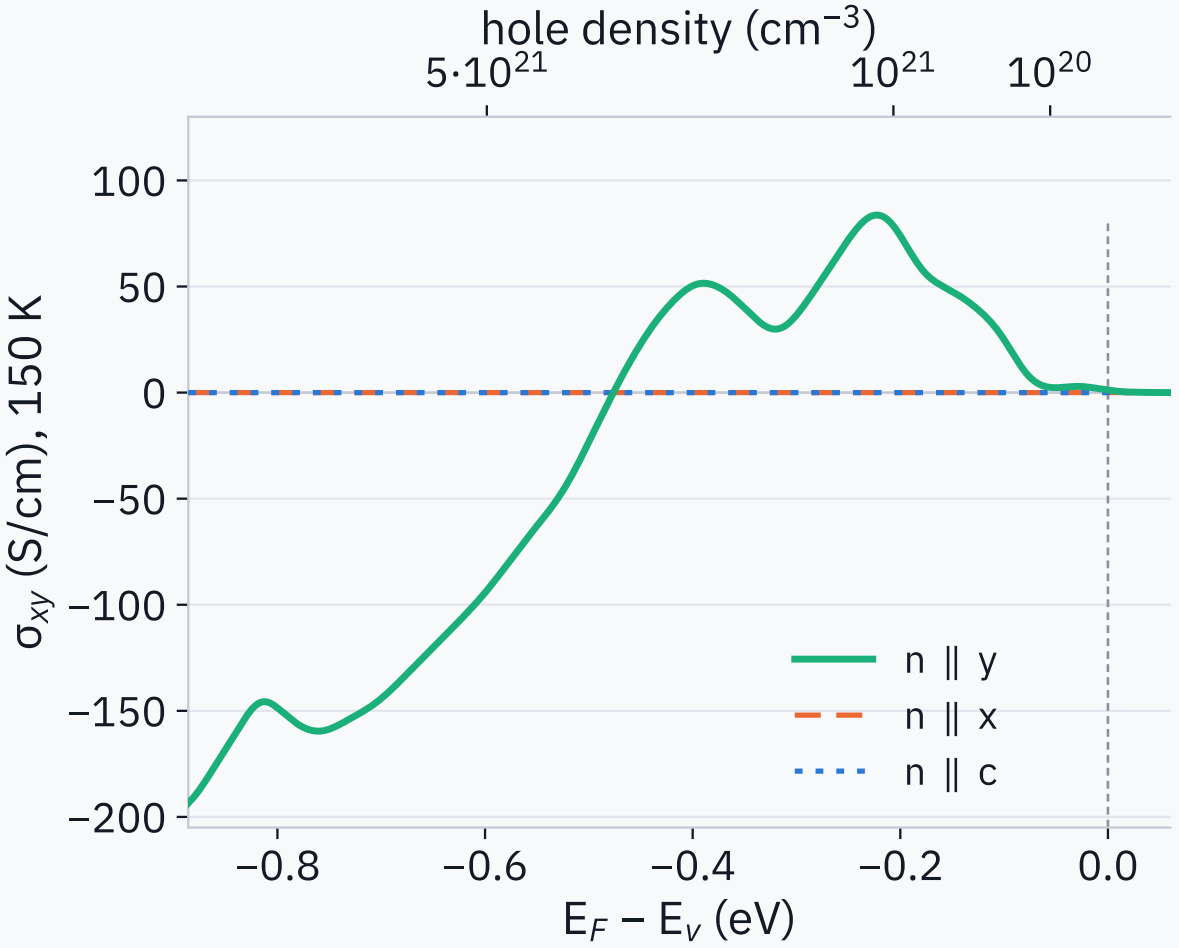


n̂ ∥ y, colour: spin along **n̂**

Anomalous Hall conductivity for three Néel vectors

```
ahc = wb.calculators.static.AHC(Efermi=Efermi,
    degen_thresh=1e-4, smoother=smoother)
directions = {"x": (90, 0), "y": (90, 90),
              "z": (0, 0)}   # theta, phi
grids = {"x": (48, 48, 32), "y": (96, 96, 64),
         "z": (48, 48, 32)}
for name, (theta, phi) in directions.items():
    system.set_soc_axis(theta=theta, phi=phi,
                        units="degrees")
    result = wb.run(system,
        grid=wb.Grid(system, NK=grids[name]),
        calculators={"ahc": ahc})
```

- $\hat{\mathbf{n}} \parallel x$ and $\hat{\mathbf{n}} \parallel c$: zero, as symmetry says
- $\hat{\mathbf{n}} \parallel y$: up to ≈ 80 S/cm, changes sign ≈ 0.5 eV below the valence band top E_V



Symmetry gives sin 3φ

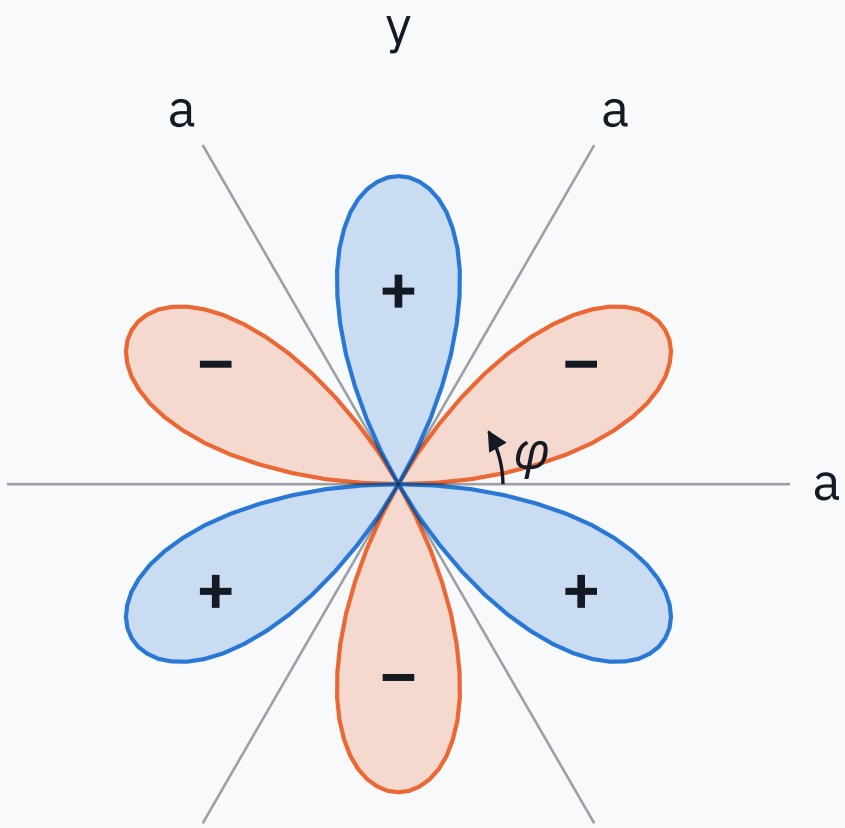
In-plane Néel vector at angle φ from a:

Operation	Swaps Mn	φ →	So
turn 180° about a	no	−φ	σ(−φ) = −σ(φ)
turn 60° about c, shift c/2, then $\mathcal{T}$	yes	φ + 60°	σ(φ + 60°) = −σ(φ)

$$\sigma_{xy}(\varphi) = A_3 \sin 3\varphi + A_9 \sin 9\varphi + A_{15} \sin 15\varphi + \ldots$$

A_m: amplitudes in S/cm

- Fixed by symmetry: zeros along every a axis, sign change every 60°
- Not fixed: the A_m; sin 3φ need not dominate



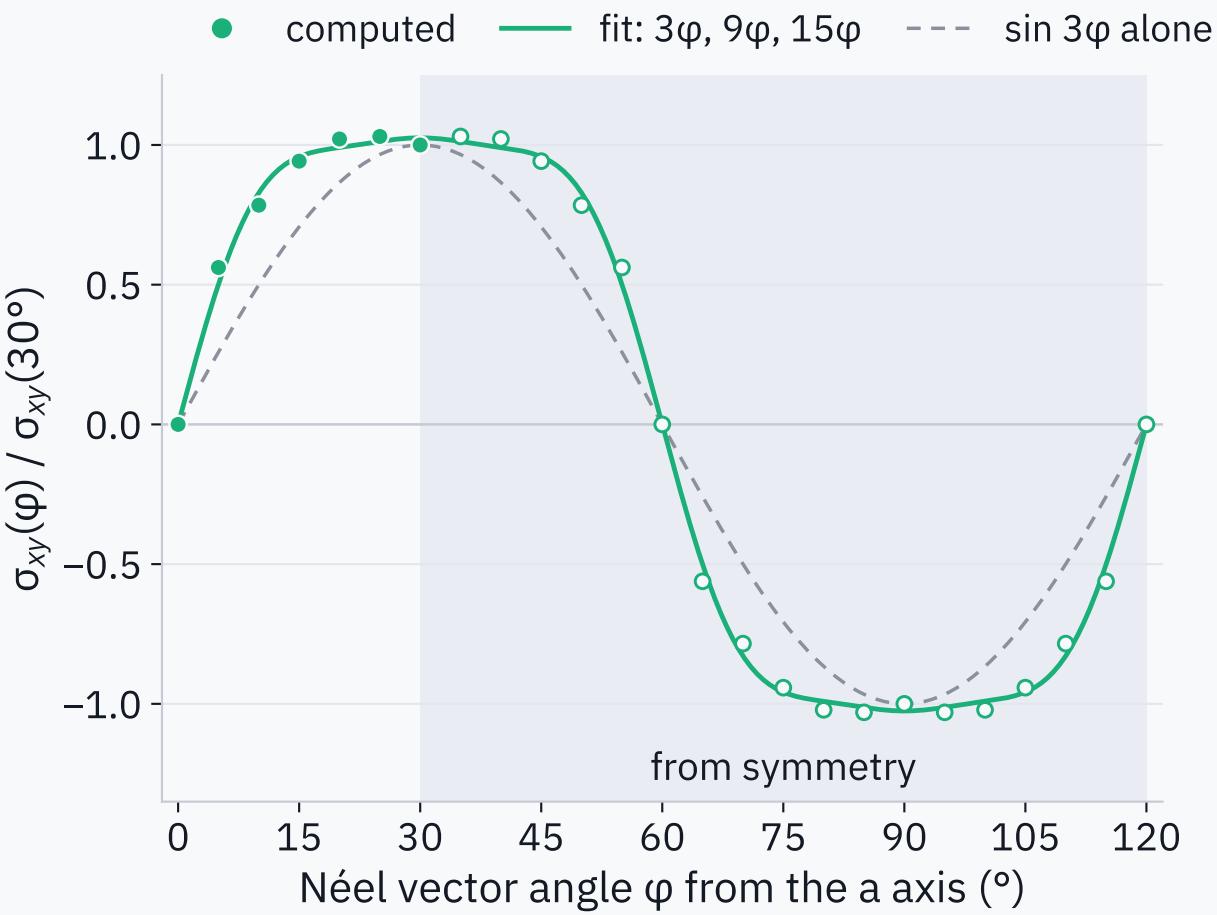
Sign of σ_{xy} against the direction of **n̂**, sin 3φ only (sketch)

Rotating the Néel vector: exact zeros, but not a sin 3φ curve

One `set_soc_axis` per angle, no new DFT: 0–30° in 5° steps, 1 min

- σ(−φ) = −σ(φ) and σ(φ + 60°) = −σ(φ) hold exactly
- Rises much faster than sin 3φ: σ(5°) is already half of σ(30°)
- σ ≈ A_3 (sin 3φ + 0.18 sin 9φ + 0.04 sin 15φ)

Strong hole doping: E_F 0.7 eV below the band top



Filled: computed. Open: from the two symmetry relations

Your material next

```
calc = GPAW(...)                                  # SCF
kpts = SpaceGroup.from_gpaw(calc)\
           .get_irreducible_kpoints_grid(grid)
calc.fixed_density(kpts=kpts, nbands=...)  # NSCF
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=...)
wandata.wannierise(froz_min=..., froz_max=...,
                   sitesym=True)
system = wb.SystemSOC.from_wannierdata(
    wandata, berry=True)
system.set_soc_axis(theta, phi)
wb.run(system, grid=..., calculators=...)
```

- Projections: the orbitals that make the bands you care about
- Frozen window: every state you need kept exactly
- k-grids: multiples of the DFT grid
- `degen_thresh` below the smallest splitting
- Check a few levels against GPAW first

SUMMARY

Take-aways, and how to run it

- Symmetry: DFT on 7 of 36 and 24 of 144 k-points
- SOC on demand: any spin axis and strength, no new DFT
- Spin Hall plateau e/2π at the real gap
- Altermagnets: Wannierise one spin channel, get both
- The Néel vector is just a parameter: symmetry gives the zeros, the bands give the size

RUN IT

github.com/yaroslav-zhumagulov/wannier-berri-demo

```
bash setup/setup.sh    # once, 15 min
conda activate wbdemo
bash run_all.sh        # 6 min, or the notebooks
```

Linux, macOS or WSL2, with GPAW 26.7 from conda-forge

BACKUP

Details

Kane–Mele form · nodal planes of MnTe · uniform grids · timings · references

The Wannier SOC matrix has the Kane–Mele form

$$H_{\mathrm{SO}} = i\lambda \sum_{\langle\langle ij \rangle\rangle} \nu_{ij}\, c_i^{\dagger} s_z c_j, \qquad \Delta_{\mathrm{SO}} = 6\sqrt{3}\,\lambda$$

- Second neighbours: **$\lambda \approx 3.4$ μeV**, purely imaginary, sign ±1 as in Kane–Mele
- $6\sqrt{3}\,\lambda \approx 36$ μeV: the whole gap of the full model
- $\Delta V^{x,y} = 0$: the mirror plane forbids Rashba terms, s_z is conserved
- No SOC between first neighbours: each bond lies in a vertical mirror

Shell	$\lvert d \rvert$ (Å)	$\lvert t \rvert$ (eV)	$\lvert \Delta V^{x,y} \rvert$ (eV)	$\lvert \Delta V^{z} \rvert$ (μeV)
1st, A–B	1.42	2.90	0	0
2nd, A–A	2.46	0.25	0	**3.4**
3rd, A–B	2.84	0.26	0	0
4th, A–B	3.76	0.02	0	0
5th, A–A	4.26	0.04	0	0
6th, A–A	4.92	0.03	0	0.2

Largest element per shell of $H(\mathbf{R})$ and $\Delta V(\mathbf{R})$; nothing fitted

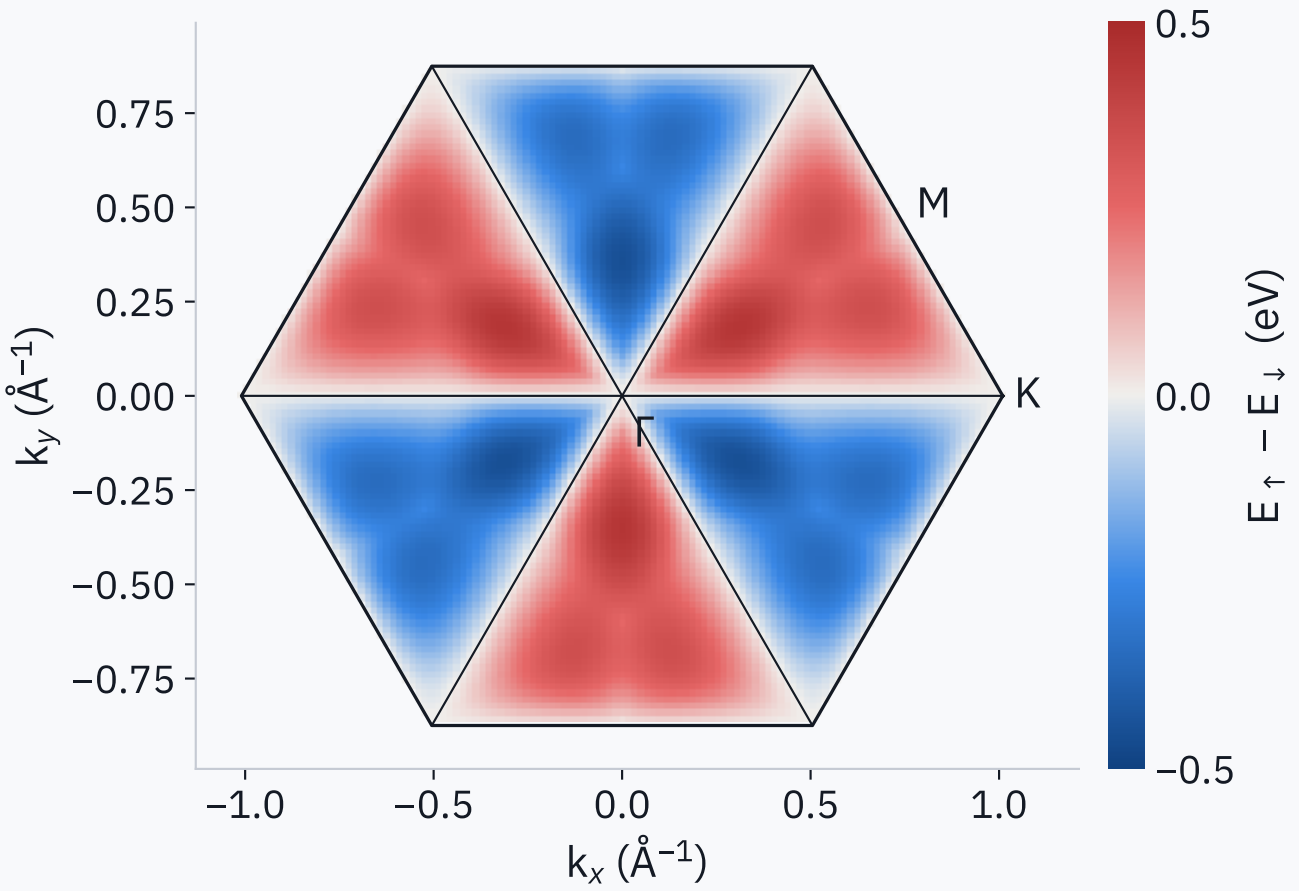
Kane, Mele, PRL 95, 226801 (2005)

Four nodal planes make MnTe g-wave

$E_\uparrow(\mathbf{k}) = E_\downarrow(R_g\mathbf{k})$ for a spin-flipping g, so the splitting vanishes on planes that R_g leaves in place:

- m_z: $k_z = 0$ and $k_z = \pi/c$
- three c-glides: three vertical planes through Γ–K

In the model the splitting is exactly zero on these planes, and up to ≈ 1 eV along Γ–M at $k_z = 0.35$ Å^{-1}



$E_\uparrow - E_\downarrow$, top valence band, $k_z = 0.35$ Å^{-1}

BACKUP

Where the time goes

Step	Graphene	MnTe
GPAW SCF + NSCF, one core	10 s	45 s
`WannierDataSOC.from_gpaw`	3.3 s	9.7 s
`wannierise`	2.1 s	9.7 s
`SystemSOC.from_wannierdata`	1.0 s	6.6 s
Conductivities (step 5)	28 s	63 s
Néel scan (step 6)	–	66 s
Animation frames	1 s	22 s
Whole notebook	1.1 min	4.2 min

Apple M2, 8 cores, 8 Ray workers: 6 minutes for everything, try-its and bonus steps included

With the Wannier model in hand, a new spin axis, SOC strength or Fermi level costs seconds

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